# Chapter 10:Gibbs Free energy Composition and Phase Diagrams of Binary Systems

# Part I

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# 10.1. Introduction

- ✓ We have seen in Chapter 7 that when temperature and pressure are the independent variables of a one-component system, the phase(s) with the lowest Gibbs free energy is (are) the stable phase(s). Likewise, in a two-component (binary) system at constant temperature and pressure, the stable state of existence at each composition of a system is that which has the lowest possible value of Gibbs free energy. Thus, phase stability, in a system presented on an isobaric temperature versus composition binary phase diagram, can be determined from knowledge of the variations of the Gibbs free energies of the various possible phases with composition and temperature.

# 10.1. Introduction

✓ When a liquid solution is cooled, a liquidus temperature is eventually reached, at which point a solid phase begins to separate from the liquid solution. This solid phase could be a virtually pure component, a solid solution of the same or different composition from the liquid, or a chemical compound formed by reaction between two or more of the components. In each of these cases, the composition of the solid phase which is in equilibrium with the liquid solution is that which minimizes the Gibbs free energy of the system.

# 10.1. Introduction

- ✓ If liquid solutions are stable over the entire range of composition, then the Gibbs free energies of the liquid states are lower than those of any possible solid-state phase. Conversely, if the temperature of the system is lower than the lowest solidus temperature, then the Gibbs free energies of the solid states are everywhere lower than those of liquid-state phases.
- ✓ At intermediate temperatures, the variation of Gibbs free energy with composition will consist of ranges of composition over which liquid states are stable, ranges over which solid states are stable, and intermediate ranges in which solid and liquid phases coexist in equilibrium with one another.

# 10.1. Introduction

- ✓ Thus, there must exist a quantitative correspondence between Gibbs free energy–composition diagrams and the equilibrium phase diagrams by virtue of the following facts:
- ✓ 1. The state of lowest Gibbs free energy is the stable state.
- ✓ 2. When phases coexist in equilibrium, the partial molar Gibbs free energy of component i, Gi , has the same value in all of the coexisting phases.
- ✓ This correspondence is examined in this chapter, in which it will be seen that temperature–composition phase diagrams are generated by, and are representations of, Gibbs free energy–composition–temperature diagrams.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

- ✓ The Gibbs free energy of mixing of the components A and B to form a mole of solution is given by

$$\Delta G^{M} = RT(X_A \ln a_A + X_B \ln a_B)$$

where $\Delta G^M$ is the difference between the Gibbs free energy of a mole of the homogeneous solution and the Gibbs free energy of the corresponding numbers of moles of the unmixed components. Since only changes in Gibbs free energy can be measured, the Gibbs free energies of the pure unmixed components are assigned the value of zero.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

- ✓ If the solution is ideal (i.e., if $a_i = X_i$ ), then the molar ideal Gibbs free energy of mixing is given by

$$\Delta G^{M,id} = RT(X_A \ln X_A + X_B \ln X_B)$$

- ✓ and has the characteristic shape shown, at the temperature *T,* as curve I in Figure 10.1. Since $\Delta H^{M,id} = 0$, then $\Delta G^{M,id} = -T\Delta S^{M,id}$. The curve I in Figure 10.1 is obtained by inverting the ideal entropy of mixing curve (Figure 9.7) and multiplying it by the temperature in question. It is thus seen that the shape of the variation of $\Delta G^{M,id}$ with composition depends only on the temperature of the solution.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

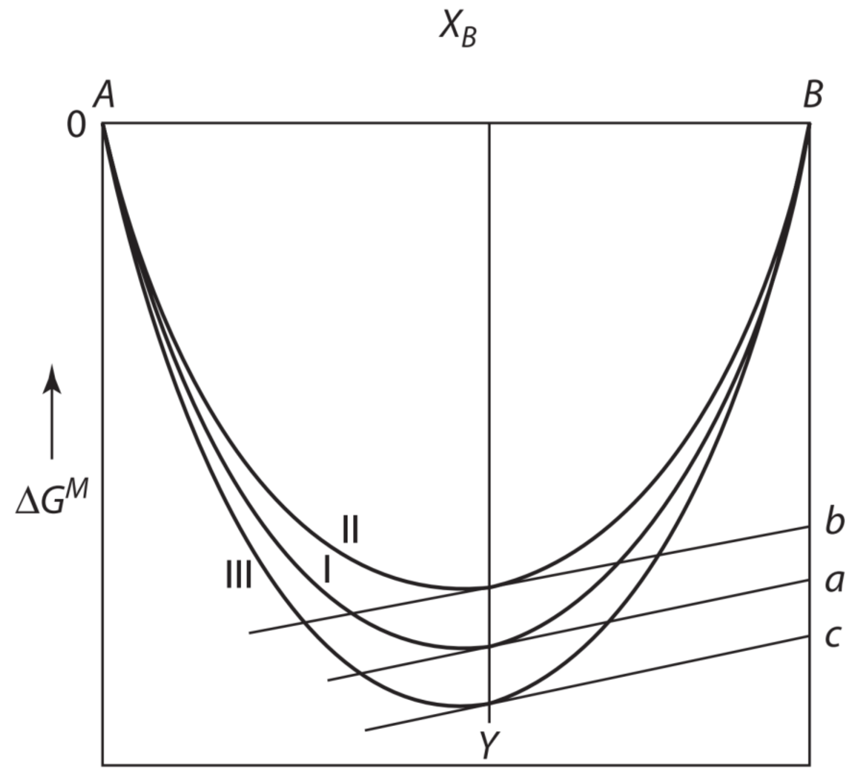


**Figure 10.1** The molar Gibbs free energies of mixing in binary systems exhibiting ideal behavior (I), positive deviation from ideal behavior (II), and negative deviation from ideal behavior (III).

# 10.2. Gibbs Free Energy And Thermodynamic Activity

✓ If the solution exhibits a small positive deviation from ideal mixing (i.e., if $\gamma_i$ >1 and $a_i > X_i$), then, at the temperature T, the Gibbs free energy of mixing curve is typically as shown by curve II in Figure 10.1. If the solution shows a slight negative deviation from ideal mixing (i.e., if $\gamma_i$ < 1 and $a_i < X_i$), the Gibbs free energy of mixing curve is typically as shown by curve III in Figure 10.1. We know from the discussion of Equation 9.33a and b that the tangent drawn to the $\Delta G^M$ curve at any composition intersects the $X_A = 1$ and $X_B = 1$ axes at $\Delta \bar{G}_A^M$ and $\Delta \bar{G}_B^M$, respectively. Also, since $\Delta \bar{G}_i^M = RT\ln a_i$, there is a relationship between the $\Delta G^M$–composition and the activity-composition plots.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

- ✓ In Figure 10.1, at the composition Y, tangents drawn to curves I, II, and III intersect the $X_B = 1$ axis at a, b, and c, respectively. Thus,

$$\left| Bb = \Delta \bar{G}_B^M = RT \ln a_B (\text{in system II}) \right| < \left| Ba = \Delta \bar{G}_B^M = RT \ln X_B \right|$$
$$< \left| Bc = \Delta \bar{G}_B^M = RT \ln a_B (\text{in system III}) \right|$$

- ✓ from which it is seen that

$$\gamma_B \text{ in system II} > 1 > \gamma_B \text{ in system III}$$

- ✓ The variation, with composition, of the tangential intercepts generates the variations of activity with composition shown in Figure 10.2. It can be seen from Figure 10.2 that the solutions II and III follow Henry's law for $X_B \longrightarrow 0$ and Raoult's law for $X_B \longrightarrow 1$.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

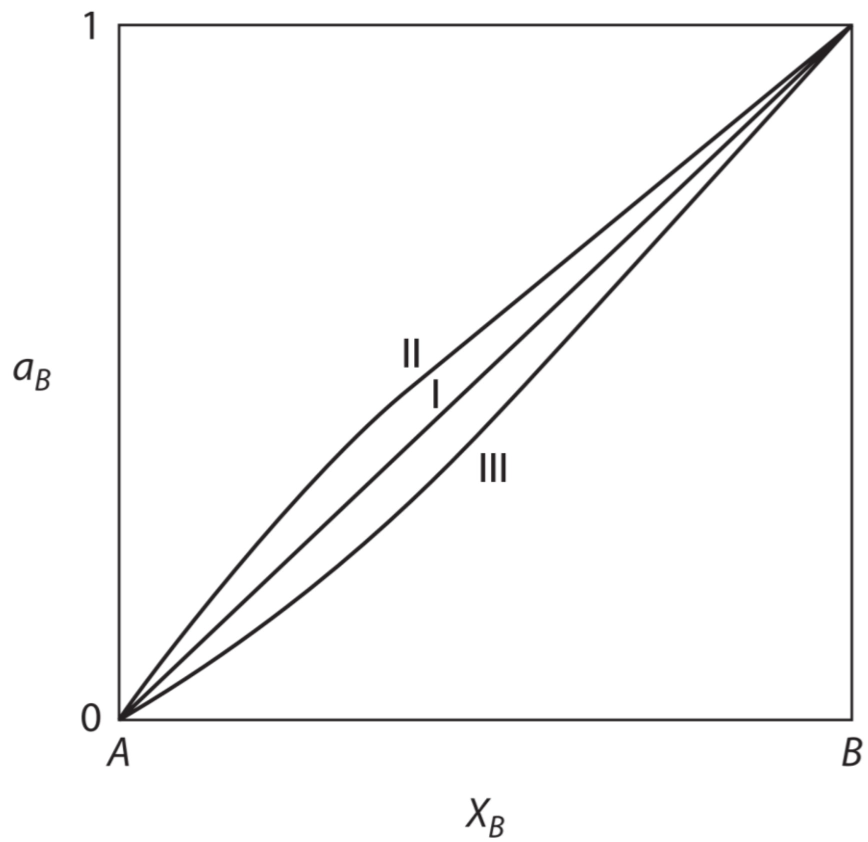


**Figure 10.2** The activities of component *B* obtained from lines I, II, and III in Figure 10.1. Solution II exhibits positive deviation from ideality, and solution III exhibits negative deviation from ideality.

# 10.2. Gibbs Free Energy And Thermodynamic Activity

- ✓ As $X_{\text{i}} \rightarrow 0$, $a_i \rightarrow 0$, the tangential intercept $\Delta\bar{G}_i^M = RT\ln a_i \rightarrow -\infty$ (see Figure10.1), which indicates that all Gibbs free energy of mixing curves have vertical tangents at their extremities.

- ✓ These vertical tangents indicate that the first atom of B added to pure A decreases the Gibbs free energy! This also can be seen from the ideal entropy of mixing curve in Figure 9.7, which by virtue of being logarithmic, has vertical tangents at its extremities. This thermodynamic principle thus implies that pure elemental components are highly unlikely.

# 10.3 Qualitative Overview of Common Binary Equilibrium Phase Diagrams

- ✓ In this section, we will examine the qualitative features of common binary equilibrium phase diagrams. We will examine the phase diagrams and discuss some of their important features. We also will look at several of the isobaric molar Gibbs free energy versus composition curves that are thermodynamically consistent with the phase diagrams. A more quantitative approach will be given in subsequent sections of this chapter.

# 10.3.1 The Lens Diagram: Regular Solution Model

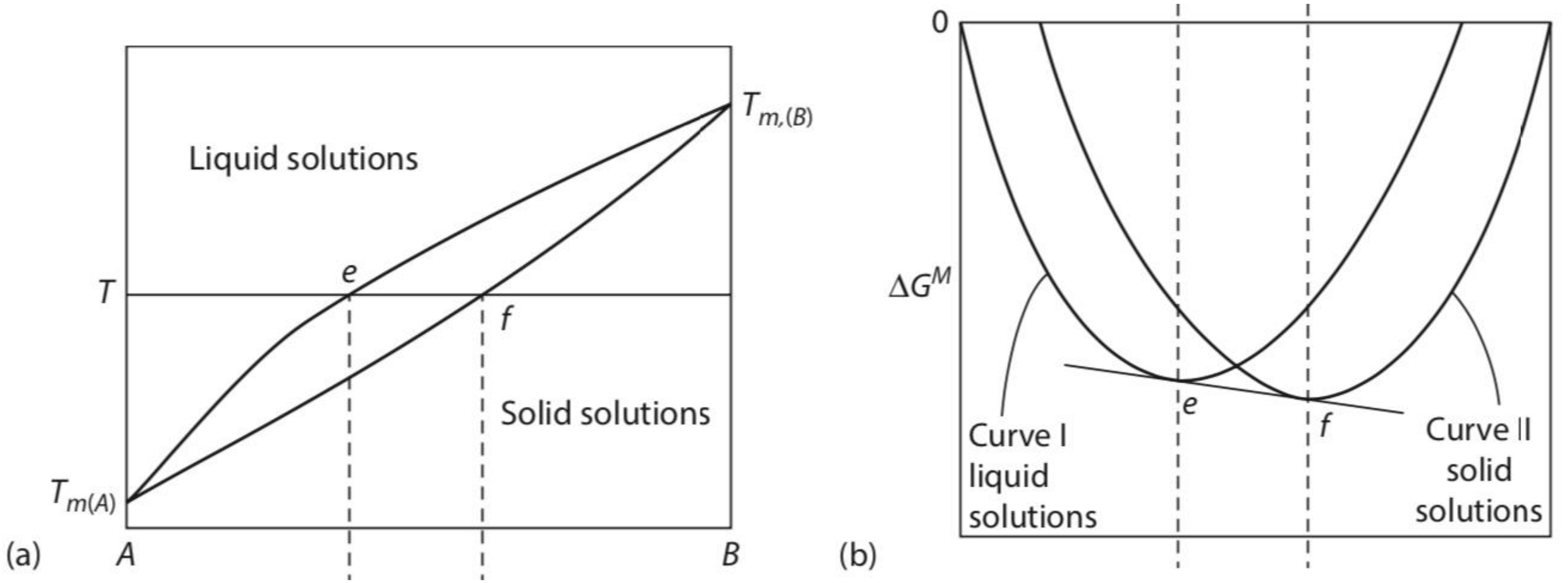


**Figure 10.3** (a) the equilibrium phase diagram for a system *A–B* exhibiting a lens diagram for the liquid–solid region. (b) the Gibbs free energies of mixing in the system *A–B* at the temperature *T*, showing the equilibrium compositions of the phases.

# 10.3.1 The Lens Diagram: Regular Solution Model

- ✓ **Figure 10.3** shows a simple way that a liquid and solid in a binary isobaric system can be in equilibrium. The end points of the diagram represent the melting points of *A*, $T_{m\,(A)}$ and of *B*, $T_{m\,(B)}$. At temperatures in between the melting temperatures, there are three equilibrium states across the composition range:
- ✓ 1. *A* liquid solution of *B* in *A*, *A (B)*
- ✓ 2. *A* two-phase mixture of liquid *A(B)* and solid *B(A)*
- ✓ 3. *A* solid solution of *A* in *B*, *B(A)*
- ✓ The lever rule determines the amount of the phases in the two-phase region. See **Section 1.7.**

# 10.3.1 The Lens Diagram: Regular Solution Model

✓ If we model the Gibbs free energies of the solid and liquid as regular solutions, and if the enthalpies of mixing are approximately the same, the Gibbs free energy terms take the form shown in **Figure 10.3b**. It can be seen that for compositions *B* < *e*, the liquid phase has the lowest Gibbs energy. For compositions of *B* > *f*, the solid phase has the lowest Gibbs energy. In between these values, the lowest Gibbs energy is obtained by a two-phase equilibrium between a liquid of composition e and a solid of composition *f*. The common tangent demonstrates that the partial molar Gibbs free energies of the two phases of each of the components are equal in the two-phase region; that is,

$$\Delta\overline{G}_{\mathrm{A}}^{\mathrm{L}} = \Delta\overline{G}_{\mathrm{A}}^{\mathrm{S}} \qquad \Delta\overline{G}_{\mathrm{B}}^{\mathrm{L}} = \Delta\overline{G}_{\mathrm{B}}^{\mathrm{S}}$$

# 10.3.1 The Lens Diagram: Regular Solution Model

- ✓ where *S* stands for solid solution and *L* stands for liquid solution. These values are found where the common tangent intersects the molar Gibbs free energy of mixing ordinates $X_B = 0$ and $X_B = 1$, respectively.
- ✓ With a change in the temperature, the Gibbs energy curves shift relative to one another, since their temperature derivatives are proportional to the entropy of the phases. As the temperature decreases, the intersections e and f shift to lower values of $X_B$ until, at $T_{m(A)}$, the Gibbs energy curve of the solid is completely below the liquid free energy curve. From that temperature and lower, only a single solid-solution phase is present.

# 10.3.2 Unequal Enthalpies of Mixing

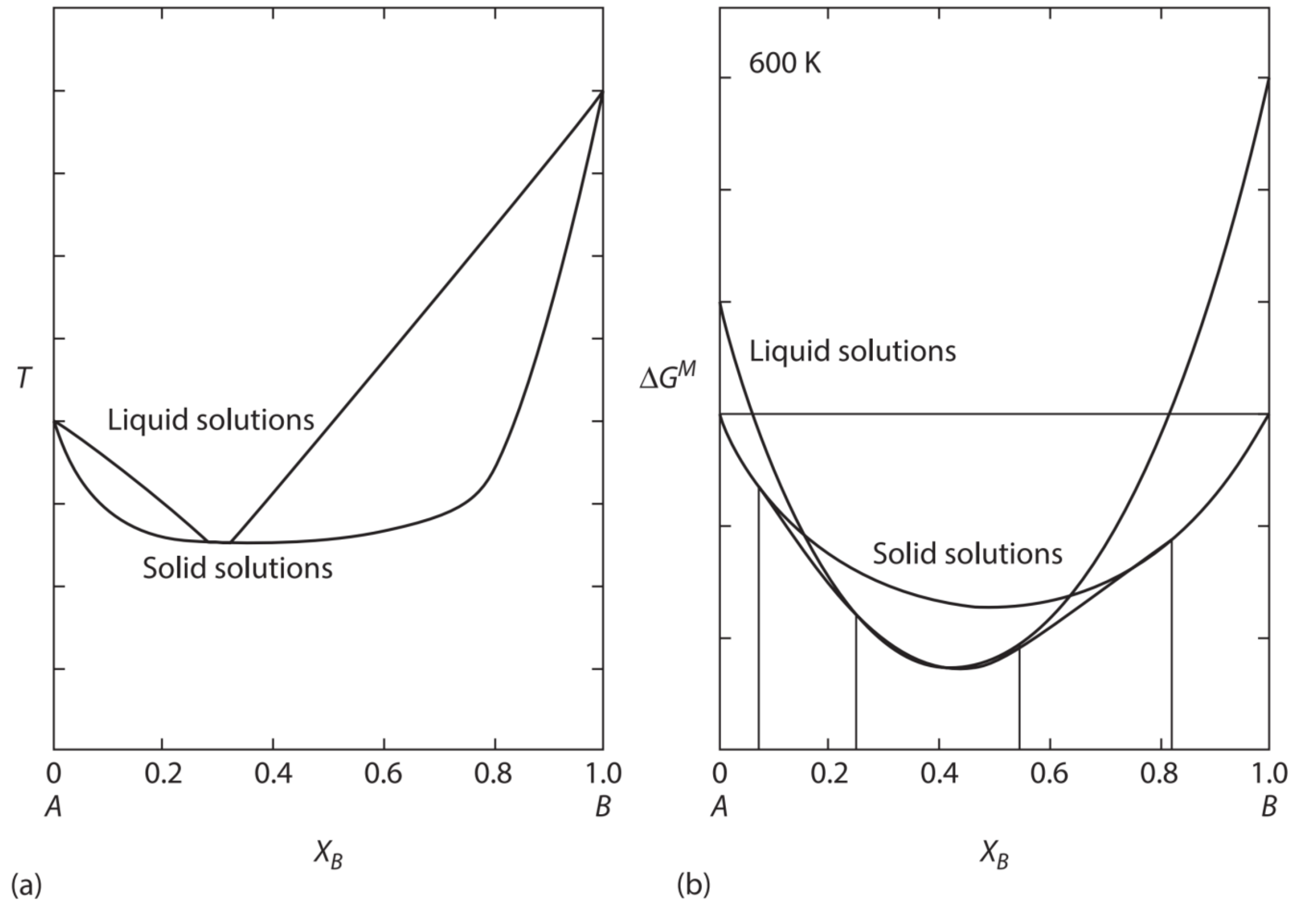


**Figure 10.4** (a) The equilibrium phase diagram of a binary system with negative enthalpy of mixing of the liquid solution and positive enthalpy of mixing of the solid solution. (b) Schematic representations of the Gibbs free energy of mixing for the solutions at the temperature $T$.

# 10.3.2 Unequal Enthalpies of Mixing

- ✓ If we model the Gibbs free energies of the solid and liquid as regular solutions, and if the enthalpy of mixing of the liquid is negative and the enthalpy of mixing of the solid is positive, the Gibbs energy curves of the phases will look as shown in **Figure 10.4b** and the phase diagram as shown in **Figure 10.4a**. The lower value of the enthalpy of mixing stabilizes the liquid solution and results in an enlarged region of stability for it. At a temperature above the congruent temperature (where the liquidus and solidus curves are tangent to each other), there are two solid–liquid equilibrium- phase fields and a single liquid-phase field in between.

# 10.3.2 Unequal Enthalpies of Mixing

- ✓ If the opposite relationship between the enthalpies of mixing were to hold (negative enthalpy of mixing of the solid and zero or positive enthalpy of mixing for the liquid), the enlarged single solid-phase field would produce a congruent point at a maximum, enlarging the field of stability of the solid solution.

- ✓ Congruent points are interesting from the point of view of the liquid-to-solid transformation in that the composition of the new phase (solid on cooling, liquid on heating) is the same as that phase from which it forms. The slopes of the solidus and liquidus curves must be zero at the congruent points and must be tangential to one another, according to the Gibbs–Konovalov rule* (Dmitry Petrovich Konovalov, 1856–1929).

- ✓ At the unique eutectic composition, this phase diagram displays the solidification reaction

$$L \rightarrow \alpha + \beta$$

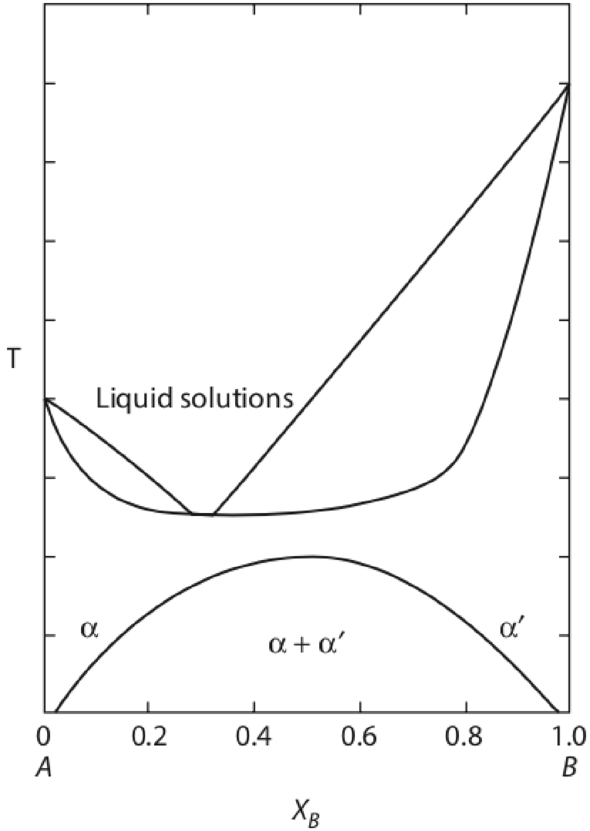


**Figure 10.5**

low-temperature region of the phase diagram shown in figure 10.4a, showing a solid-state miscibility gap.

# 10.3.3 The Low-temperature Regions in Phase Diagrams

✓ In **Figures 10.3** and **10.4**, once the temperature is below the regions of liquid stability, a single solid solution is seen to be present. This solid solution may exist down to room temperature and even to lower temperatures, but cannot continue as a single-phase solid solution region to very low temperatures because of atomic interactions (see **Section 6.5** on the Third Law of Thermodynamics). Since the enthalpy of mixing of the solid was said to be positive for the system shown in **Figure 10.4**, the solution will exhibit a low-temperature miscibility gap, as shown in **Figure 10.5**. This is consistent with the Third Law, in that at 0 K, the two phases in equilibrium would be pure *A* and *B*, and therefore have zero configurational entropy. The free energy curves that develop such miscibility gaps for the solid solution will be discussed in **Section 10.5.**

# 10.3.4 The Eutectic And Eutectoid Phase Diagrams

- ✓ Another frequently encountered binary phase diagram type is shown in **Figure 10.6,** denoted as the eutectic phase diagram (from the Greek for “easy melting”). In this isobaric phase diagram, there are three equilibrium phases: two solid phases ($\alpha$ and $\beta$) and a liquid phase. Above the invariant temperature (the horizontal line, known as the eutectic temperature), the diagram is divided into regions similar to those displayed in **Figure 10.4a**—that is, two single-phase solid regions (one of $\alpha$ and the other of $\beta$), two two-phase regions ($\alpha/L$ and $\beta/L$), and a single-phase liquid region.

# 10.3.4 The Eutectic And Eutectoid Phase Diagrams

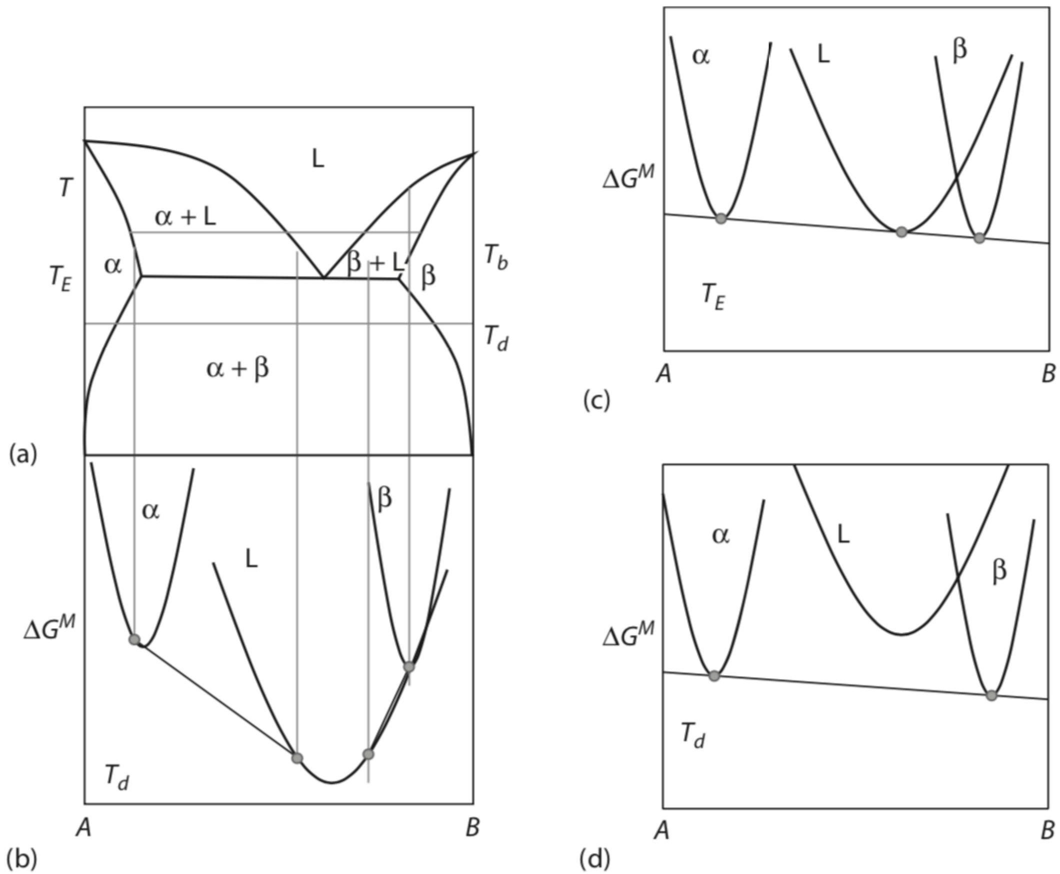


**Figure 10.6** (a) an isobaric eutectic binary phase diagram displaying two solid phases (*α* and *β*) and one liquid phase. (b) Gibbs free energy of mixing curves of the three phases at a temperature above the eutectic temperature. (c) Gibbs free energy curves of the three phases at the eutectic temperature $T_E$ . (d) Gibbs free energy curves of the three phases at the temperature $T_d$.

- ✓ which is commonly called a eutectic reaction. Two solid-phase solutions form from the liquid solution concomitantly, often giving rise to striking microstructures. Use of the Gibbs equilibrium phase rule shows that in a binary isobaric system, the three phases *α, L*, and *β* can be in equilibrium with each other only at a fixed temperature (i.e., the equilibrium has zero degrees of freedom). Therefore, the three-phase equilibrium must exist on a horizontal line (fixed temperature) in isobaric binary phase diagrams.
- ✓ In this diagram, there are two other temperature-invariant reactions that may occur involving three phases: namely, the hypoeutectic reaction

$$\alpha + \mathrm{L} \rightarrow \alpha + \beta$$

## 10.3.4 The Eutectic And Eutectoid Phase Diagrams

- ✓ and the hypereutectic reaction

$$\beta + L \rightarrow \alpha + \beta$$

- ✓ The free energy of mixing curves at three temperatures for this phase diagram are shown in **Figure 10.6 b,c, and d. Figure 10.6b** shows the relative positions of the three free energy of mixing curves at a temperature above the eutectic temperature $T_E$. Note that the liquid free energy curve lies between those of the two solids. Also, it can be seen that above the eutectic temperature, $T_E$, the free energy of mixing of the liquid lies below those of the two solids.

# 10.3.4 The Eutectic And Eutectoid Phase Diagrams

- ✓ As the temperature decreases, the free energy of mixing curve of the liquid rises faster than those of the solids. This occurs because the temperature dependence of the free energy, $((\partial G/\partial T)_P = -S)$, is the negative of the entropy of the phase, and since the liquid has the larger entropy, it will be displaced to higher values of free energy of mixing at a greater rate than the two solid curves. At the eutectic temperature, $T_E$, the three free energy of mixing curves of the phase lie on a common tangent (**Figure 10.6c**).

- ✓ Thus, all three phases have equal partial molar Gibbs free energies; that is,

$$\Delta\bar{G}_A^L = \Delta\bar{G}_A^\alpha = \Delta\bar{G}_A^\beta$$

$$\Delta\bar{G}_B^L = \Delta\bar{G}_B^\alpha = \Delta\bar{G}_B^\beta$$

# 10.3.4 The Eutectic And Eutectoid Phase Diagrams

- ✓ Below $T_E$, only the two solid phases are present in equilibrium and their compositions continuously shift to that of the pure components *A* and *B*.
- ✓ If the high-temperature phase in these diagrams is replaced by a solid phase, the diagram is called a eutectoid (eutectic-like) phase diagram. The alloy system *Fe–C* has the important constant-temperature eutectoid transformation $\gamma \rightarrow \alpha + Fe_3C$, producing the well-known constituent pearlite (*a* two-phase lamellar mixture of $\alpha + Fe_3C$).

# 10.3.5 The Peritectic And Peritectoid Phase Diagrams

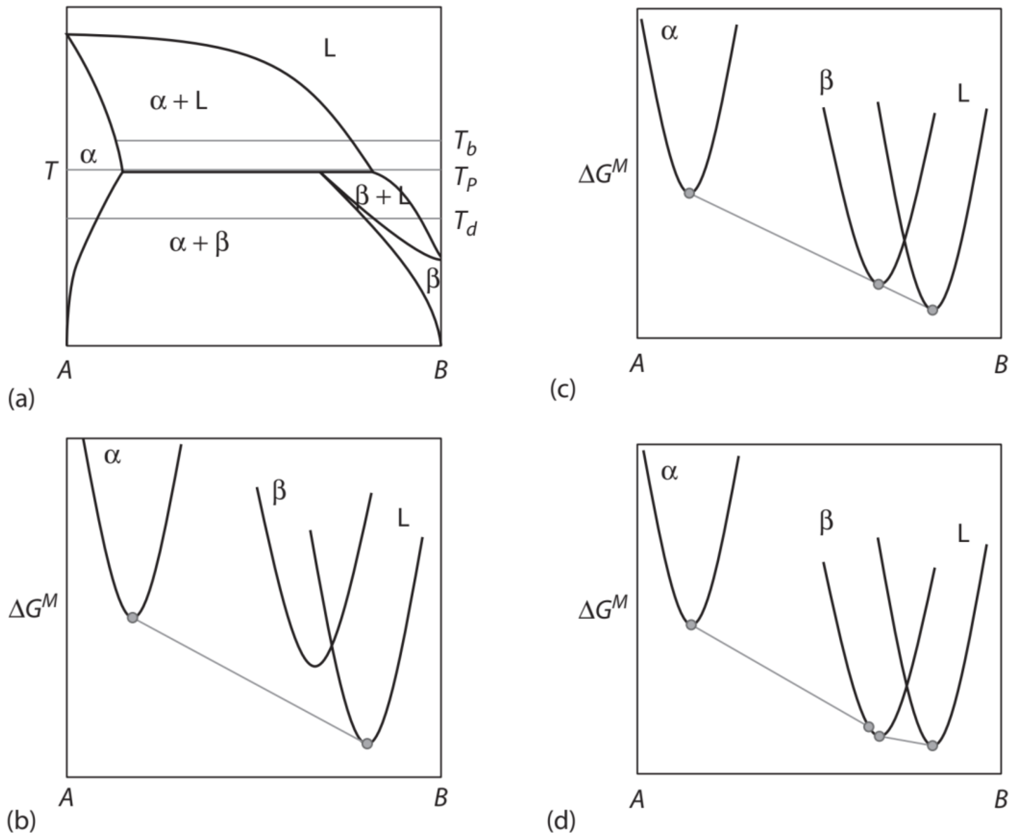


**Figure 10.7** (a) A peritectic binary phase diagram displaying two solid phases (a and b) and one liquid phase. (b) Gibbs free energy of mixing curves of the three phases at a temperature above the peritectic invariant temperature. Note that the free energy curve of the liquid phase is to the *B*-rich side of both free energy curves of the solid phases. (c) Gibbs free energy of mixing curves of the three phases at the peritectic temperature $T_P$. (d) Gibbs free energy of mixing curves of the three phases at the temperature $T_d$.

# 10.3.5 The Peritectic And Peritectoid Phase Diagrams

- ✓ In the eutectic phase diagrams, the high-temperature liquid phase has a composition between that of the two low-temperature phases at the invariant temperature. It is possible for the liquid phase to be the richest in solute (or solvent) at the invariant temperature, as shown in **Figure 10.7**. For these diagrams, commonly called peritectic diagrams, on cooling, an alloy of the peritectic composition passing through the invariant temperature undergoes a transformation of the form

$$\alpha + L \rightarrow \beta$$

  - ✓ This is followed by the reaction

$$\beta \rightarrow \alpha + \beta$$

# 10.3.5 The Peritectic And Peritectoid Phase Diagrams

- ✓ There are two other invariant reactions that may occur involving three phases in the peritectic system: namely, the hypo-peritectic reaction

$$\alpha + L \rightarrow \alpha + \beta$$

- ✓ And the hyper-peritectic reaction

$$\alpha + L \rightarrow \beta + L$$

- ✓ The arrangements of the Gibbs free energy of mixing curves necessary to obtain such a phase diagram are shown in **Figure 10.7b** and c. It can be seen that the free energy of mixing curve of the liquid lies to the solute-rich side of both the solid phases in the diagram.

## 10.3.5 The Peritectic And Peritectoid Phase Diagrams

- ✓ If the liquid phase is replaced by a third solid phase, *α,* the diagram will look similar but without the liquid phase. Such a diagram is called a peritectoid diagram. In this case, for an alloy of the peritectoid composition cooling from the *α* + *γ* region, the invariant reaction is

$$\alpha + \gamma \rightarrow \beta$$

- ✓ There are other invariant types of binary equilibrium phase diagrams. For example, **Figures 10.23 and 10.26** display monotectic phase diagrams in which the invariant reaction is

$$L_1 \rightarrow \alpha + L_2$$

- ✓ Also, a metatectic (or catatectic) diagram displays the invariant reaction on cooling

$$\beta \rightarrow \alpha + L$$

# 10.4 Liquid And Solid Standard States

- ✓ The choice of the standard state of a component of a condensed system is often that of the pure component in its stable state at the particular temperature and pressure of interest. At 1atm pressure (the pressure normally considered), the stable state is determined by whether or not the temperature of interest is above or below the normal melting temperature of the component. It is often assumed that the temperature of interest is either above or below the melting temperatures of both components.

# 10.4 Liquid And Solid Standard States

- ✓ An activity versus composition curve could be drawn for liquid immiscibility, in which case the standard states are the two pure liquids, or it could be drawn for solid immiscibility, in which case the standard states are the two pure solids. Since the standard state of a component is simply a reference state with which the component in any other state is compared, it follows that any state can be chosen as the standard state, and the choice is normally made purely on the basis of convenience.

- ✓ Consider the binary system *A*–*B* at the temperature T, which is below the melting temperature of *B*, $T_{m(B)}$, and above the melting temperature of *A*, $T_{m(A)}$. Consider, further, that this system forms Raoultian ideal liquid solutions and Raoultian ideal solid solutions.

# 10.4 Liquid And Solid Standard States

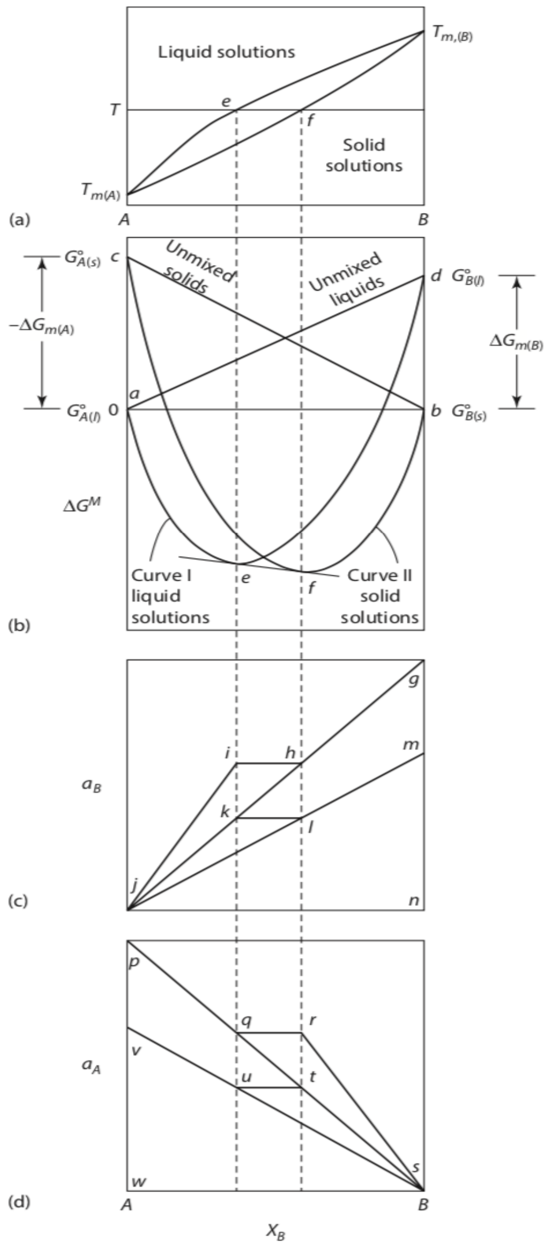


**Figure 10.8**

(a) the phase diagram for the system *A*–*B*. (b) the Gibbs free energies of mix- ing in the system *A*–*B* at the temperature *T*. (c) the activities of *B* at the temperature *T* and comparison of the solid and liquid standard states. (d) the activities of *A* at the temperature *T* and comparison of the solid and liquid standard states.

# 10.4 Liquid And Solid Standard States

- ✓ The phase diagram for the system is shown in **Figure 10.8a. Figure 10.8 b** shows the two Gibbs free energy of mixing curves at the temperature of inter- est. Curve I is drawn for liquid solutions and curve II is drawn for solid solutions.

- ✓ At the temperature *T*, the stable states of pure *A* and *B* are located at $\Delta G_M = 0$, with pure liquid *A* located at $X_A = 1$ (the point a) and pure solid *B* located at $X_B = 1$ (the point *b*). The point c represents the molar Gibbs free energy of solid *A* relative to that of liquid *A* at the temperature *T* , and $T > T_{m(A)}$.Thus $G^o_{A(s)}-G^o_{A(l)}$ is a positive quantity which is equal to the negative of the molar Gibbs free energy of melting of *A* at the temperature *T.* That is,

$$G^{\circ}_{A(s)} - G^{\circ}_{A(l)} = -\Delta G^{\circ}_{m(A)} = -\left(\Delta H^{\circ}_{m(A)} - T\Delta S^{\circ}_{m(A)}\right)$$

# 10.4 Liquid And Solid Standard States

- ✓ and if $c_{PA(s)} = c_{PA(l)}$ (i.e., if $\Delta H_{m(A)}$ and $\Delta S_{m(A)}$ are independent of temperature), then

$$\Delta G^{\circ}_{m(A)} = \Delta H^{\circ}_{m(A)} \left(\frac{T_{m(A)} - T}{T_{m(A)}}\right)$$

- ✓ As the temperature decreases and approaches the melting point of *A*, $\Delta G^{o}_{m(A)}(T)$ approaches zero.

- ✓ Similarly, the point d represents the molar Gibbs free energy of liquid B relative to that of solid *B* at the temperature *T*, and, since $T < T_{m(B)}$, then $G^{o}_{B(s)} - G^{o}_{B(l)}$ is a positive quantity equal to $\Delta G^{o}_{m(B)}$ .

# 10.4 Liquid And Solid Standard States

- ✓ The line in **Figure 10.8a** joining a and d represents the Gibbs free energy of unmixed liquid *A* and liquid *B* relative to that of the standard state of unmixed liquid *A* and solid *B*, and the line joining *c* and *b* represents the Gibbs free energy of unmixed solid *A* and solid *B* relative to that of the standard state. The line *cb* can be represented by the equation

$$\Delta G = -X_A \Delta G^{\circ}_{m(A)}$$

- ✓ and the equation for the line *ad* is

$$\Delta G = X_B \Delta G^{\circ}_{m(B)}$$

- ✓ At any composition, the formation of a homogeneous liquid solution from pure liquid

# 10.4 Liquid And Solid Standard States

- ✓ A and pure solid *B* can be considered as being a two-step process involving
- ✓ 1. The melting of $X_B$ moles of *B*, which involves the change in Gibbs free energy
- ✓ 2. The mixing of $X_B$ moles of liquid *B* and $X_A$ moles of liquid *A* to form an ideal liquid solution, which involves the change in Gibbs free energy

$$\Delta G = \Delta G^{M,id} = RT(X_A \ln X_A + X_B \ln X_B)$$

# 10.4 Liquid And Solid Standard States

- Thus, the molar Gibbs free energy of mixing (which can also be called the molar Gibbs free energy of formation) of an ideal liquid solution, $\Delta G^{M}_{(l)}$ , from liquid *A* and solid *B* is given by

$$\Delta G^{M}_{(l)} = RT(X_A \ln X_A + X_B \ln X_B) + X_B \Delta G^{\circ}_{m(B)}$$

- which is the equation of curve I in **Figure 10.8b**.
- Similarly, at any composition, the formation of an ideal solid solution from liquid *A* and solid *B* involves a change in Gibbs free energy of

$$\Delta G^{M}_{(s)} = RT(X_A \ln X_A + X_B \ln X_B) - X_A \Delta G^{\circ}_{m(A)}$$

- which is the equation of curve II in **Figure 10.8b.**

# 10.4 Liquid And Solid Standard States

- ✓ At the composition *e*, the tangent to the free energy of mixing curve for the liquid solution is also the tangent to the free energy of mixing curve of the solid solution at the composition *f*. Thus, at the temperature *T*, the liquid of composition *e* is in equilibrium with the solid of composition *f*; that is, *e* is the liquidus composition and *f* is the solidus composition, as seen in **Figure 10.8a**.

- ✓ As the temperature is lowered, the magnitude of *ca* decreases until, at the melting point of *A*, the length of *ca* equals zero. With decreasing temperature, the magnitude of *db* increases. The consequent movement of the positions of curves I and II relative to one another is such that the positions *e* and *f* of the double tangent to the curves shift to the left.

# 10.4 Liquid And Solid Standard States

- ✓ Correspondingly, if the temperature is increased, the relative movement of the Gibbs free energy curves is such that *e* and *f* shift to the right. The loci of *e* and *f* with change in temperature trace out the liquidus and solidus curves, respectively.

- ✓ The equations for the solidus and liquidus curves of this system are derived in Appendix 10A of this chapter. There, the equations are found to be

$$X_{A(s)} = \frac{1 - \exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})}{\exp(-\frac{\Delta G^{\circ}_{m(A)}}{RT}) - \exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})}$$

# 10.4 Liquid And Solid Standard States

and

$$X_{A(l)} = \frac{\left[1 - \exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})\right] \exp(-\frac{\Delta G^{\circ}_{m(A)}}{RT})}{\exp(-\frac{\Delta G^{\circ}_{m(A)}}{RT}) - \exp(-\frac{\Delta G^{\circ}_{m(B)}}{RT})}$$

- ✓ If $c_{p,i(s)} = c_{p,i(l)}$, we obtain from **Equation 10.1**, for $i = A$ and $B$,

$$\Delta G^{\circ}_{m(i)} = \Delta H^{\circ}_{m(i)} \left[\frac{T_{m(i)} - T}{T_{m(i)}}\right]$$

- ✓ Thus, the phase diagram for a system which forms ideal solid and liquid solutions is determined only by the melting temperatures and the molar heats of melting of the components.

# 10.4 Liquid And Solid Standard States

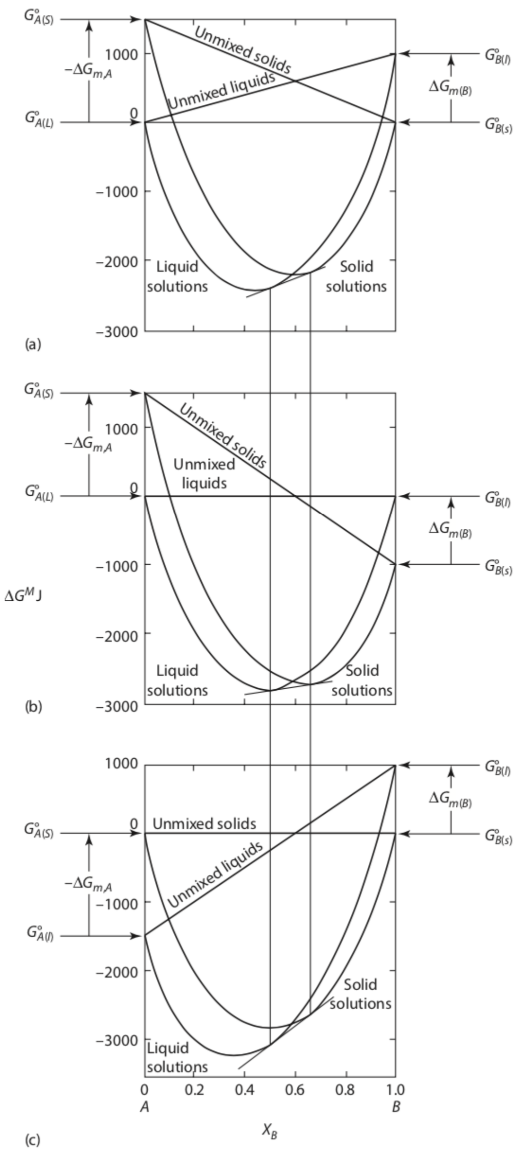

**Figure 10.9**

the Gibbs free energy of mixing curves for a binary system *A*–*B* which forms ideal solid solutions and ideal liquid solutions, at a temperature which is higher than $T_{m(A)}$ and lower than $T_{m(B)}$. (a) liquid *A* and solid *B* chosen as standard states located at $\Delta G^M = 0$. (b) liquid *A* and liquid *B* chosen as standard states located at $\Delta G^M = 0$. (c) solid *A* and solid *B* chosen as standard states located at $\Delta G^M = 0$. the positions of the points of double tangency are not influenced by the choice of standard state.

# 10.4 Liquid And Solid Standard States

✓ **Figure 10.9** shows the Gibbs free energy of mixing curves for a binary system A–B which forms ideal solid solutions and ideal liquid solutions, drawn at a temperature of 500 K, which is lower than $T_{m,(B)}$ and higher than $T_{m,(A)}$. At 500 K, $\Delta G^{o}_{m,A} = -1500\ J$ and $\Delta G^{o}_{m,B} = 1000\ J$. **Figure 10.9a** shows the curves when liquid *A* and solid *B* are chosen as the standard states, located at $\Delta G^M = 0$. **Figure 10.9b** shows the curves when liquid *A* and liquid *B* are chosen as the standard states, and **Figure 10.9c** shows the curves when solid *A* and solid *B* are chosen as the standard states.

# 10.4 Liquid And Solid Standard States

- ✓ Comparison among the three shows that, because of the logarithmic nature of the Gibbs free energy curves, the positions of the points of common tangency are not influenced by the choice of standard state; they are determined only by the temperature *T* and by the magnitude of the difference between $G^o_{(l)}$ and $G^o_{(s)}$ for both components at the temperature *T*.

# 10.4 Liquid And Solid Standard States

✓ The activity–composition relationships for component *B* are shown in **Figure 10.8c.** There are two possible standard states (**Figure 10.8b**). The point *b* for solid *B* and the point *d* for liquid *B* could be chosen as standard states. The lengths of the tangential intercepts with the $X_B = 1$ axis can be measured from *b*, in which case the activities of *B* are obtained with respect to solid *B* as the standard state, or the lengths can be measured from *d*, which gives the activities with respect to liquid *B* as the standard state.

# 10.4 Liquid And Solid Standard States

- ✓ If pure solid *B* is chosen as the standard state and is located at the point *g* in **Figure 10.8c**, then the length *gn* is, by definition, unity, and this defines the solid standard-state activity scale. The line *ghij* then represents $a_B$ in the solutions, with respect to solid *B* having unit activity at *g*. The line is obtained from the variation of the tangential intercepts from the curve *aefb* to the $X_B = 1$ axis, measured from the point *b*. On this activity scale, Raoult' s law is given by *jg*, and the points *i* and h represent, respectively, the activity of *B* in the coexisting liquid solution *e* and solid solution *f.*

# 10.4 Liquid And Solid Standard States

- ✓ The point m represents the activity of pure liquid *B* measured on the solid standard-state activity scale of *B.* This activity is less than unity, being given by the ratio *mn/gn*. For *B* in any state along the *aefb* Gibbs free energy of mixing curve, in which state the partial molar Gibbs free energy of *B* is $\bar{G}_B$, the following relations hold.

$$\bar{G}_B = G^{\circ}_{B(l)} + RT \ln (a_B \text{ with respect to liquid } B)$$

- ✓ and

$$\bar{G}_B = G^{\circ}_{B(s)} + RT \ln (a_B \text{ with respect to liquid } B)$$

- ✓ Thus, $G^{\circ}_{B(l)} - G^{\circ}_{B(s)} = \Delta G^{\circ}_{m(B)} = RT \ln \left(\dfrac{a_B \text{ with respect to solid } B}{a_B \text{ with respect to liquid } B}\right)$

# 10.4 Liquid And Solid Standard States

- ✓ Since $T < T_{m(B)}$ , $\Delta G^o_{m(B)}$ is a positive quantity, the activity of $B$ in any solution with respect to solid $B$ as the standard state is less than the activity of $B$ with respect to liquid $B$ as the standard state, where both activities are measured on the same (solid or liquid) activity scale. For pure $B$, $a_B\ (s) > a_B\ (l)$ (*i.e.*, $gn > mn$ in **Figure 10.3c**), and, if $gn = 1$, then $mn = exp(-\Delta G^o_{m(B)}/RT\ )$. **Equation 10.6** simply states that the length of the tangential intercept from any point on the curve *aefb*, measured from *b* added to the length *bd* equals the length of the tangential intercept from the same point on the curve measured from *d*, which is a restatement of **Equation 10.6**.

# 10.4 Liquid And Solid Standard States

- ✓ If pure liquid *B* is chosen as the standard state and is located at the point *m*, then the length *mn* is, by definition, unity, and this defines the liquid standard-state activity scale. Raoult’ s law on this scale is given by the line *jm*, and the activities of *B* in solution, with respect to pure liquid *B* having unit activity, are represented by the line *mlkj*. The activity of solid *B*, located at *g*, is greater than unity on the liquid standard-state activity scale, being equal to exp$(\Delta G^o m(B) / RT)$. When measured on one or the other of the two activity scales, the lines jihg and *jklm* vary in the constant ratio exp$(\Delta G^o m(B) / RT)$, but *jihg* measured on the solid standard-state activity scale is identical with *jklm* measured on the liquid standard-state activity scale.

# 10.4 Liquid And Solid Standard States

- ✓ The variation of $a_A$ with composition is shown in **Figure 10.8d**. In this case, since $T > Tm(A)$ , $\Delta G^o m(A)$ is a negative quantity, and hence,

$$a_A(\text{with respect to liquid A}) > a_A(\text{with respect to solid A})$$

- ✓ when measured on the same activity scale. If pure liquid *A* is chosen as the standard state and is located at the point *p*, then the length of *pw* is, by definition, unity, and the line pqrs represents the activity of *A* in the solution with respect to the liquid standard state. On the liquid standard-state activity scale, the activity of pure solid *A*, located at the point *v*, has the value exp $(\Delta G^o_{m(A)}/ RT)$.

# 10.4 Liquid And Solid Standard States

✓ If, on the other hand, pure solid *A* is chosen as the standard state, then the length of vw is, by definition, unity, and Raoult's law is given by versus. The line vuts represents the activities of A in the solutions with respect to pure solid *A*. On the solid standard-state activity scale, liquid A, located at the point p, has the value exp ($-\Delta G^{o}_{m(A)}/RT$). Again, the two lines, measured on one or the other of the two activity scales, vary in the constant ratio exp ($\Delta G^{o}_{m(A)}/RT$), and when measured on their respective scales, they are identical.

# 10.4 Liquid And Solid Standard States

✓ If the temperature of the system is decreased to a value less than $T$ indicated in **Figure 10.8a**, then the length of ac, being equal to $|\Delta G^{o}_{m(A)}|$ at the temperature of interest, decreases; correspondingly, the magnitude of $|\Delta G^{o}_{m(B)}|$ , and hence the length of *bd*, increases. The consequent change in the positions of the Gibbs free energy of mixing curves I and II in **Figure 10.8b** causes the double tangent points *e* and *f* to shift to the left toward *A*. The effect on the activities is as follows. In the case of both components,

$$\frac{a_i \text{ with respect to solid } i}{a_i \text{ with respect to liquid } i} = \exp\left(\frac{\Delta G^{\circ}_{m(i)}}{RT}\right)$$

# 10.4 Liquid And Solid Standard States

$$\frac{a_i \text{ with respect to solid } i}{a_i \text{ with respect to liquid } i} = \exp\left(\frac{\Delta G^\circ_{m(i)}}{RT}\right)$$

which, from **Equation 10.1,**

$$= \exp\left[\Delta H^\circ_{m(i)}\left(\frac{T_{m(i)} - T}{RTT_{m(i)}}\right)\right]$$

# 10.4 Liquid And Solid Standard States

- ✓ With respect to component *B*, if the temperature, which is less than $T_{m(B)}$, is decreased, the ratio $a_{B(solid)}/a_{B(liquid)}$, which is greater than unity, increases. Thus, in **Figure 10.8c**, the ratio *gn/mn* increases. With respect to the component *A*, if the temperature, which is higher than $T_{m(A)}$, is decreased, then the ratio $a_{A(solid)}/a_{A(liquid)}$, which is less than unity, increases. Thus, the ratio *vw/pw* in **Figure 10.8d** increases. At the temperature $T_{m(B)}$, solid and liquid *B* coexist in equilibrium, $\Delta G^o_{m(A)} = 0$ , and the points *p* and *v* coincide. Similarly, at the temperature $T^o_{m(B)}$ , the points *m* and *g* coincide.

# Homework

1. $CaF_2$ and $MgF_2$ are mutually insoluble in the solid state and form a simple binary eutectic system. Calculate the composition and temperature of the eutectic melt assuming that the liquid solutions are Raoultian. The actual eutectic occurs at $X_{CaF_2}$ =0.45 and T = 1243 K.

2. Gold and silicon are mutually insoluble in the solid state and form a eutec-tic system with at eutectic temperature of 636 K and a eutectic composition of $X_{Si}$ = 0.186. Calculate the Gibbs free energy of the eutectic melt relative to

   (a) unmixed liquid Au and liquid Si, and

   (b) unmixed solid Au and solid Si.

# Homework

3. $Na_2O.B_2O_3$ and $K_2O.B_2O_3$ form complete ranges of solid and liquid solutions and the solidus and liquidus show a common minimum at the equimolar composition and T = 1123 K. Calculate the molar Gibbs free energy of formation of the equimolar solid solution from solid $Na_2O.B_2O_3$ and solid $K_2O.B_2O_3$ at 1123 K, assuming that the liquid solutions are ideal.

# *The End*